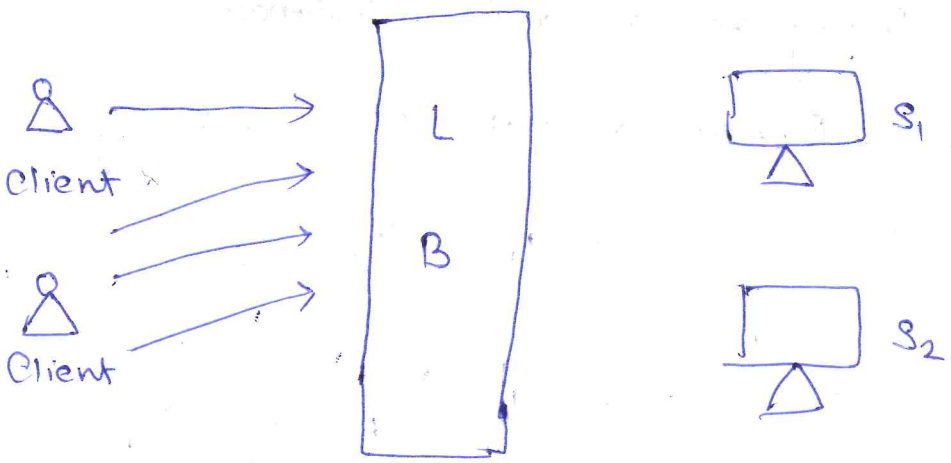
18:- Load Balancer & Different Algorithms :→



When a request comes from client. All the request not only going to one server. so one server should not get overloaded. so what we ~~put~~ do is we put Load Balancer.

Concept && Coding

Load Balancer :- The main purpose of load Balancer It can do logging. It can do so many stuff but the main purpose distribute this traffic appropriately to all the Servers so that no one single Server overburdened.

~~Application Load Balancer is work~~

OSI — 7 Application
—
—
— 4 Transport layer
—
—
—

* Application layer is known as L7 Balancer (read your header/ Session/Data/Response)
* Transport layer is known as L4 Balancer → (It can have your TCP port

Load Balancer

Network Load Balancer Type (L4 Balancer)

* It can have TCP port / UDP Port / IP Address / Source and distination so based on this only it can take ~~destination~~ desision whether to Rout this traffic to which Server.

* This is much more faster.

Application Load Balancer Type (L7 Balancer)

* It can take decision to distribute this traffic based upon information from header, Session, Cookies, data.

* It can also have capbility to do caching. Because It can read the response.

* This is much more Advanced.

Concent && Coding

Network Load Balancer Type (L4 Balancer)

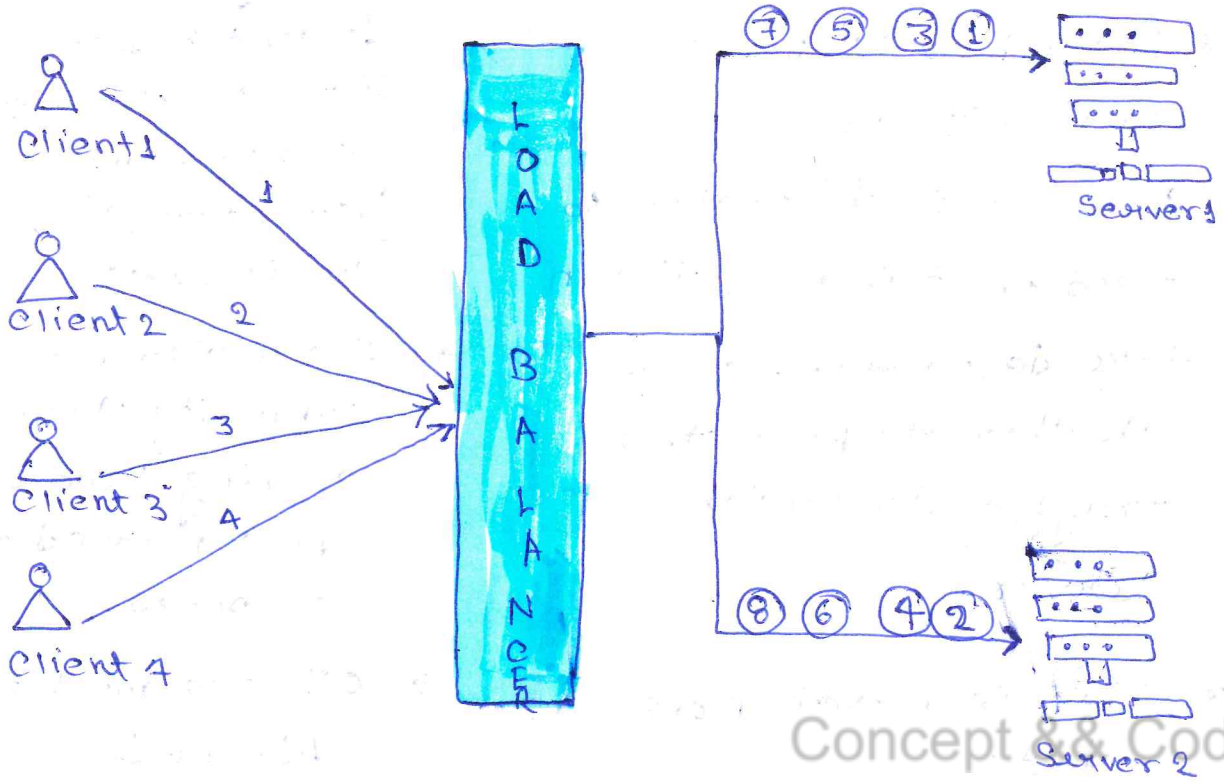
Static Algorithms

- Round Robin
- Weighted Round Robin
- Ip Hash

Dynamic Algorithms

- Least Connection
- Weighted Least Connection
- Least Response Time.

1: ROUND ROBIN



Concept && Coding

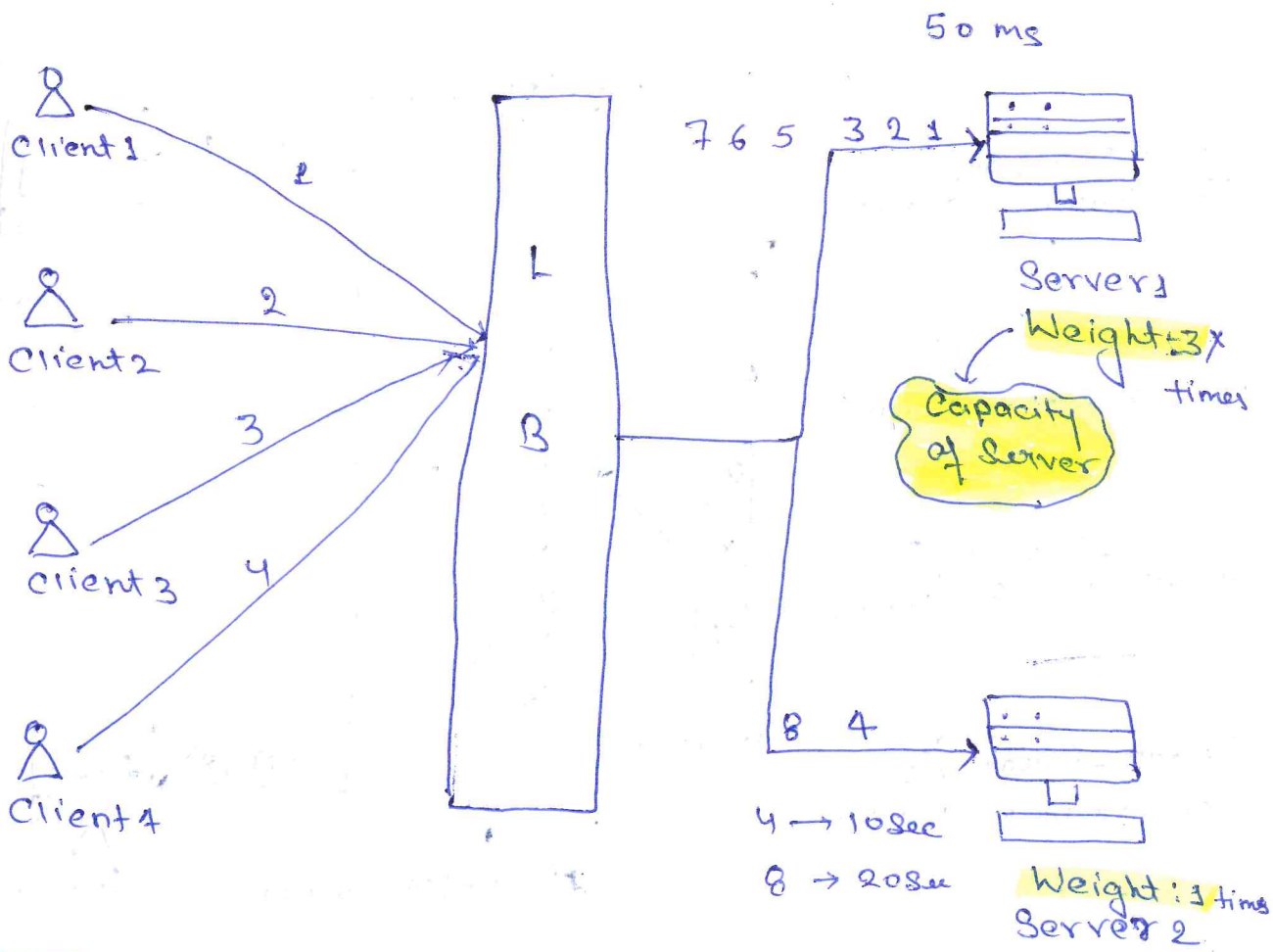
Advantages:-

- Very Easy to implement
- Equal load distributedon to all Servers.

Disadvantages:-

- One Server with high Capacity and Another with low, both will treated as same.
- Chance that low capacity Server will go down because of overload of requests.

2: WEIGHTED ROUND ROBIN :-



Concept && Coding

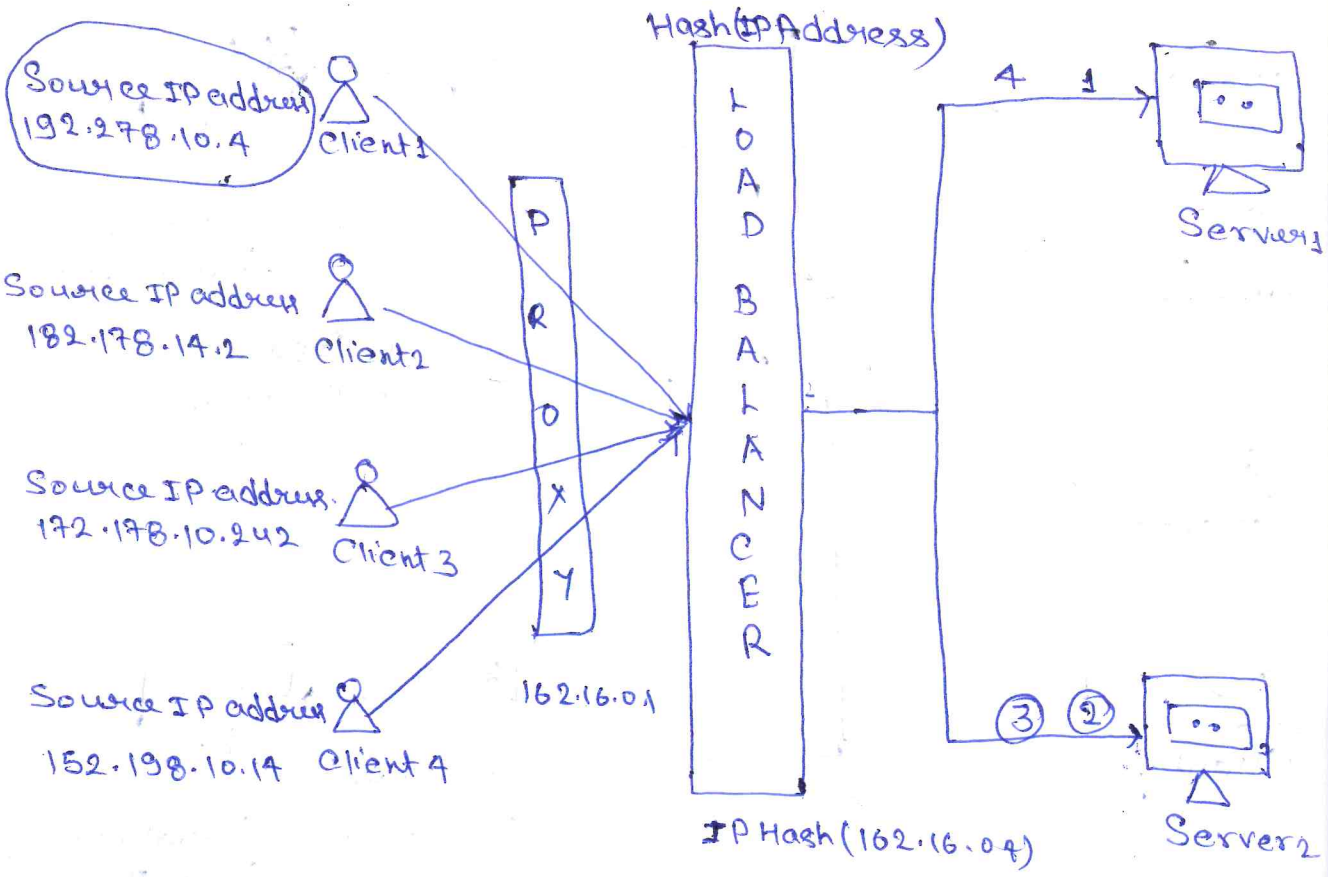
Advantages :-

- Low Capacity Server, will get saved from receiving the large no of requests.
- Easy to implement as weights are static, no dynamic computation.

Disadvantages :-

- If requests have <u>different processing time</u>, then its possible that low capacity server get high processing requests and get overburdened.

3. IP HASH :-



Concept && Coding

* Here the Load Balancer use this Source IP address to do compute the Hash(IP Address) number so using Hash function you will get some value and based upon that value you will assign the request to the server.

Advantages

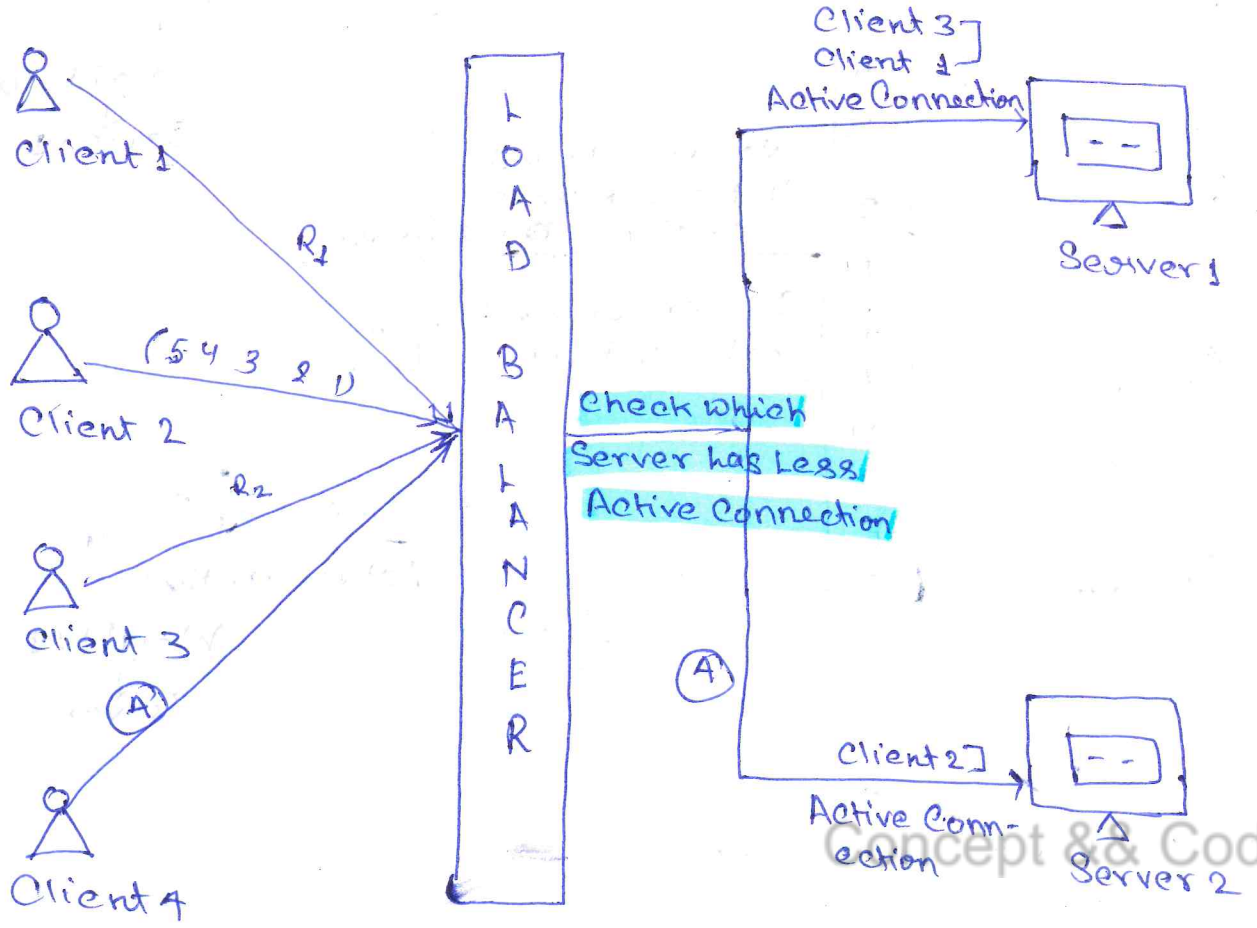
- Good for use cases, where Same client need to connect to Same Server.
- Easy to implement.

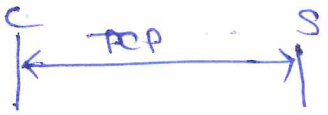
Disadvantages :-

- If Clients request is comming through PROXY, then all the clients will same source IP address, and this will OVERLOAD one server.
- Can not ensure equal distribution

DYNAMIC LOAD BALANCING

1:- LEAST CONNECTION :-





Concept && Coding

Advantages

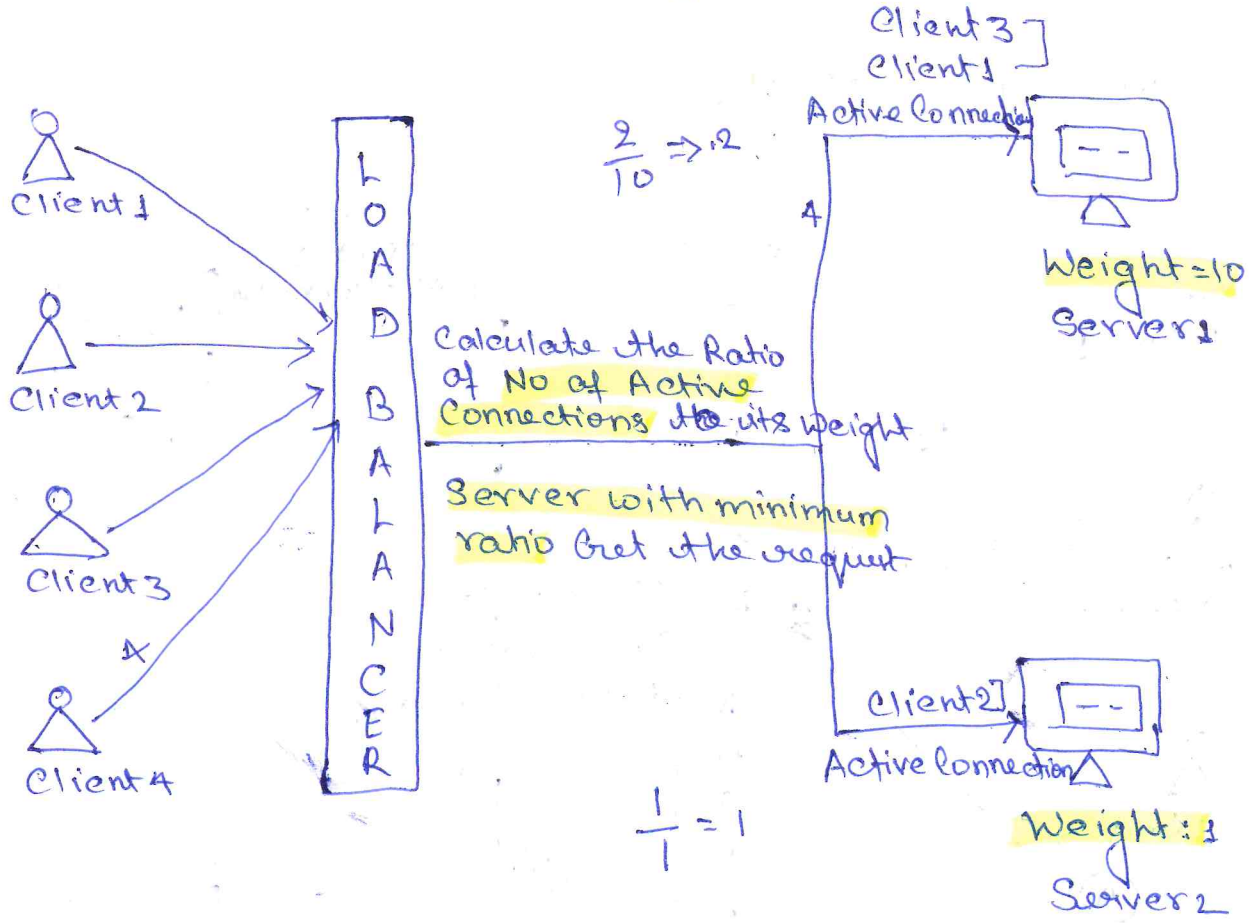
- Dynamic, as it consider the Load on each Server, so chance of overburdened of the Server is less when each server has equal capacity.

Disadvantages

- TCP connections can be ACTIVE but possible have no traffic. So purpose is failed.

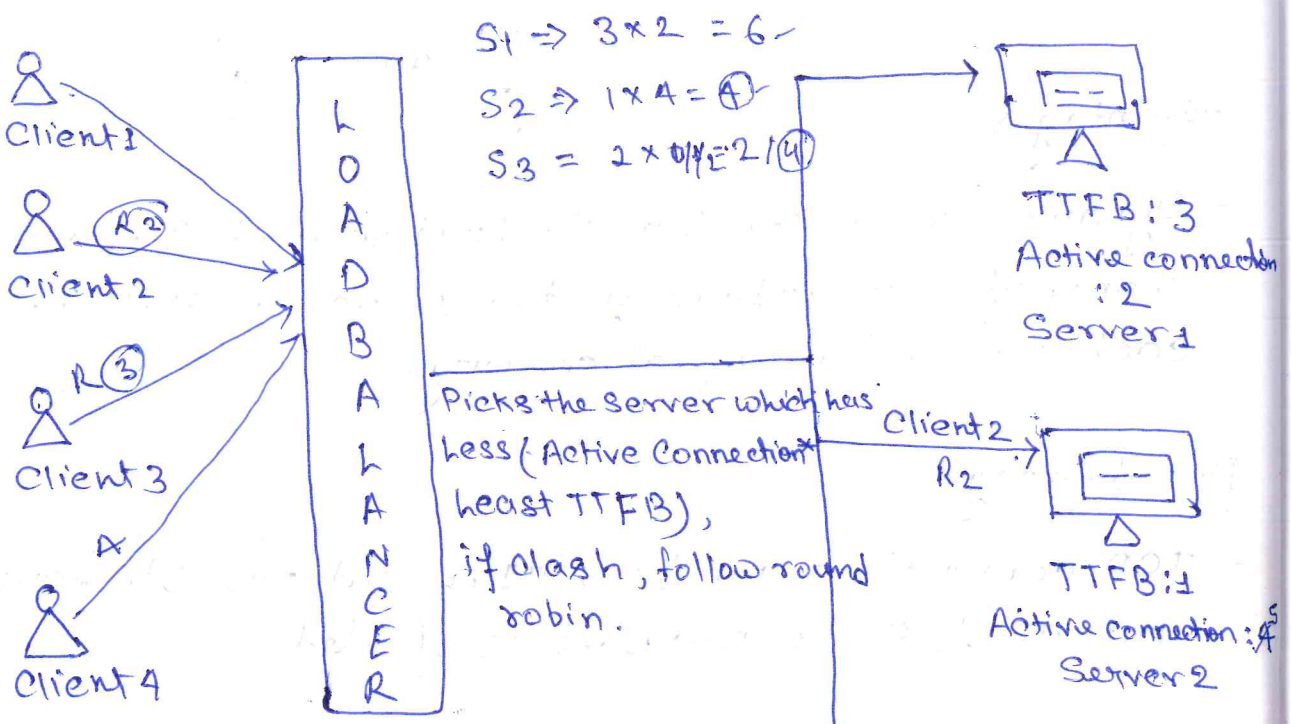
- No Difference between Low capacity and High capacity Server, chance that Low capacity Server will get over burdened.

2:- WEIGHTED LEAST CONNECTION :→



3:- LEAST RESPONSE TIME :-

Concept && Coding



TTFB (Time to First Byte)

time interval between sending a request and receiving the response from the Server.
→ first Byte